



VENKAT MATTELA
FOUNDER, CHAIRMAN AND
CHIEF EXECUTIVE OFFICER,
REDPINE SIGNALS INC.

Artificial Intelligence:

“In The Next Industrial Revolution, The COMMODITY WILL BE AI”

We have been hearing a lot about the Internet of Things (IoT) in the last few years. What comes next? Venkat Mattela, founder, chairman and chief executive officer, Redpine Signals Inc., speaks with Dilin Anand from EFW

Q. What will be the next technological evolution after the IoT?

A. Artificial Intelligence (AI) is the next technological step after the IoT. With the IoT you can control and monitor machines through the Internet, but it stops there. You still need to manually look at your phone to increase your room's temperature or open an app to turn on your smart lighting system. With AI, the machine itself can figure out when to turn on or modulate devices around you based on your previous interactions with these.

Q. What is the flip side of AI controlling computing infrastructure and machines?

A. The flip side is that, security and its issues will transform the usage drastically over the years. Today, humans are responsible for most hacks across the planet. However, the day we see AI grow past the tipping point, we will see hacks and cyber attacks happening 24/7 a day, simultaneously from billions of connected devices. New security systems will be developed to counter security threats from smart devices.

Q. How do we ensure security in connected devices?

A. Using keys for protection is very important, and protecting keys is a challenge. We have a physically un-clonable function in hardware that even its developer cannot break

into because it is based on the physics of the chip. Even if you try to copy my design, the key will not be the same.

FIPS 140-2 certification is very important for securing healthcare devices, and it is enforced through a chip. No one wants a wireless connection to be made in a hospital without any security, because there is a lot of sensitive data in there.

Q. What makes AI a tipping point in technological evolution?

A. In the next industrial revolution, the commodity will be AI. Any IoT-enabled factory or industry can be quickly connected to AI, which can then go through years of historic data to figure out trends that can drastically improve the efficiency and effectiveness of that production facility. The level of productivity increases or energy efficiency that can be brought in is unimaginable.

Q. What makes AI hard?

A. Algorithms are domain-specific. There is no single set of algorithms for all AI problems. If you take human intelligence, it is multi-dimensional. One of these could be of arithmetic type.

Computers too have an arithmetic type of thinking, with pattern recognition and so on. With a single trained dimension, a machine can beat a human because it has no distractions. With numerous wireless

options available today and applications of social networks, the amount of raw data coming into AI is incredibly high.

Q. What is a good blend of wireless technologies for IoT projects today?

A. Wi-Fi 2.4 and 5GHz, BLE 5.0, Bluetooth, Thread and a couple of other proprietary wireless technologies for provisioning purposes and 802.11p. Our new SoC, OctaCombo, has eight wireless technologies on the same die.

Then, there is DecaCombo with ten wireless technologies on the same die that has Global Navigation Satellite System as well as LTE for IoT. A modern IoT chip has a microcontroller, wireless connectivity, low power and security features—all located on the same die.

Q. Why are single-chip and platform solutions the most popular?

A. A typical IoT developer team is not like a mobile phone design and developer team. The latter has thousands of people because it intends to sell millions of units.

In the IoT, at least for now, companies hope to sell tens of thousands of units, and therefore the teams are small, so they cannot afford to work with multiple vendors. That is the reason we have platforms like WyzBee, where a developer/OEM can create a complete IoT solution. **EFW**